

Patterns in Attendance, Participation, and Cancer Type Across Member Groups

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Tutorial Group: TUT0201, Group number: 2

Introduction

Supportive therapies like yoga and tai chi have been shown to improve outcomes for cancer survivors [1]. This motivates our study of engagement in Wellspring's free cancer support programs.

To help these programs become more accessible, Wellspring launched a simplified registration system on March 4, 2024. We examine whether participation patterns changed after this system update and analyze whether age is related to participation. Then we consider breast cancer, even though breast cancer is more common among women, men can also be affected—accounting for about 0.2% of cases in Canada [2]. If most breast cancer programs are designed for women, male patients may be underserved. Thus we examine whether this gender pattern appears in Wellspring's members to help inform inclusive program design.

Research Question 1

Did the registration system change impact the proportion of missed events among new members?

This question focuses on members who joined before and after the system change, and compares their missed event proportions to see if there was a difference in attendance.

Motivation

Wellspring recently introduced a simplified registration system on March 4, 2024, with the goal of making it easier for potential members to get started and stay engaged in cancer support programs. To measure engagement, it is important to compare the proportion of events that members attend as scheduled. If the new system helped reduce confusion or administrative barriers, we would expect members who registered after the system update to miss fewer events. This research question focuses on whether the proportion of missed events among newly registered members changed after the system update.

Data Cleaning

To get the data ready, we first removed any rows with missing values in key columns (Table 1). Then, we grouped members as “new” or “old” based on when they joined: those who joined in 2023 or early 2024 were “old,” and those who joined after March 2024 were “new.” We removed March joiners because the new registration system started on March 4, 2024, but the dataset only shows the month members joined, not the exact date. So for people who joined in March, we can’t tell if they signed up before or after the change.

Table 1: Summary of Data Cleaning for Question 1

Reason for Removal	Number of Members Removed
Missing data removed	906
March 2024 members removed	195

We filtered out 906 members who didn’t have enough info and filtered out 195 members who joined in March 2024. Therefore, we removed 1101 members in total.

Attendance Differences Between Groups

We compare the missed event proportions between new and old members to see whether their attendance differed (Figure 1).

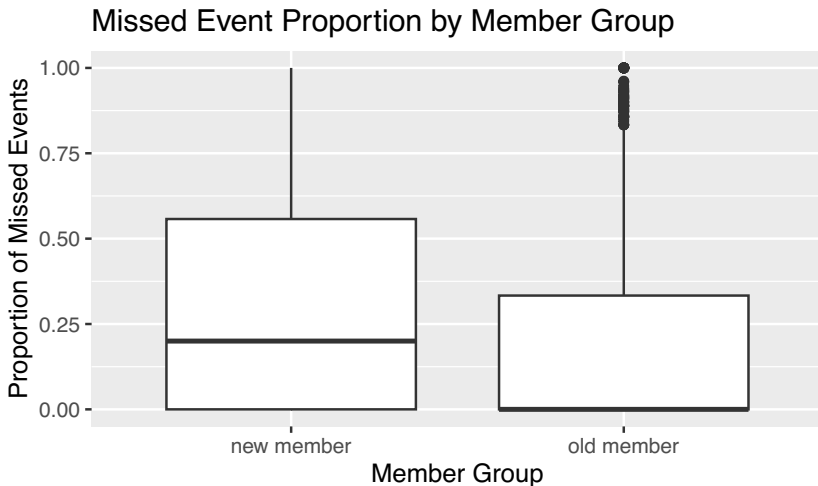


Figure 1: Missed Event Proportion by Member Group

What We Learn from the graph

- ▶ The distribution for both groups is right-skewed, suggesting that most members attended a large portion of the events they registered for.
- ▶ The median missed proportion is clearly higher for new members, while old members are more concentrated at lower proportions.
- ▶ The box for new members is longer, which means their attendance varies more—some attend most events, while others miss many.
- ▶ The box for old members is shorter, showing more consistent attendance, but they have more outliers which means some patients missed nearly all the events they signed up for.

Attendance Summary: New vs Old Members

To summarize the data, we will presents the average proportion of missed events for each member group, along with the standard deviation and sample size (Table 2).

Table 2: Missed Event Summary for New and Old Members

Member Group	Average Missed Proportion	Standard Deviation	Sample Size
new member	0.3143866	0.3509579	1046
old member	0.1868500	0.2666195	2682

- ▶ The variation is slightly higher among new members ($SD = 0.35$) compared to old members ($SD = 0.27$), meaning their attendance behavior is more inconsistent.
- ▶ The observed difference in missed event proportions between new and old members was around 12.8 percentage points(31.4% vs 18.7%), with new members missing more events on average.

Method: Hypothesis Test Comparing 2 Groups

1. We used a randomization test to compare the average missed event proportions between new and old members.
2. In each simulation, we randomly shuffled the group labels and re-calculated the difference in average missed events.

p-value: 0

- ▶ Using a two-sample hypothesis test with 1,000 permutations, we find that the proportion of missed events is higher for new members (31.4%) than for old members (18.7%) with very strong evidence ($p < 0.001$).
- ▶ This suggests that the difference is statistically significant, and not likely due to random chance, so the new system may be linked to more missed events by new members.
- ▶ One possible reason is that the simplified registration process made it easier for people to sign up without fully committing to attending and it may led to more uncertain registrations.

Research Question 2

Is there a relationship between age and participation in Wellspring support programs?

This question examines whether member age is associated with how much they participate in Wellspring support programs.

Motivation

Since our earlier findings suggest that the registration system change influenced participation, we were also curious whether age plays a role in how engaged members are. If a relationship exists — especially a linear one — Wellspring could use this insight to better design age-targeted outreach and programming strategies.

Data Cleaning

We removed members with missing age or attendance data (present or absent), as well as those with invalid age values ($\text{age} < 0$) when both attendance fields were present, so that we could calculate each member's participation rate for linear regression analysis (Table3).

Table 3: Summary of Data Cleaning for Question 2

Reason for Removal	Number of Members Removed
Missing age or attendance data	2305
Invalid age (< 0)	1

A total of 2,306 members were excluded due to missing values in age or attendance data, or due to invalid age values ($\text{age} < 0$ with no missing attendance information).

Scatterplot of Participation Rates by Age

This scatterplot shows the relationship between members' age and their participation in Wellspring programs. Each point represents one member, and the blue line shows the best-fit linear trend. (Figure 2)

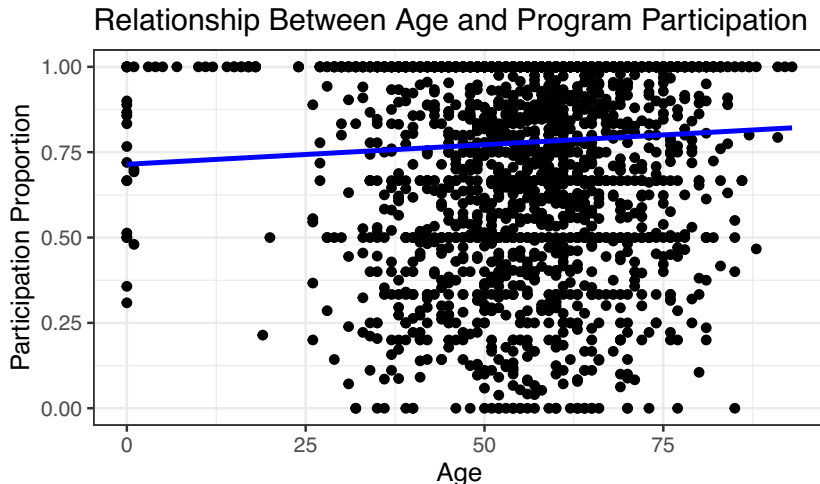


Figure 2: Relationship Between Age and Program Participation

What We Learn from the graph

- ▶ The linear trend line is slightly upward, which suggests a very weak positive relationship between age and participation.
- ▶ Many members participated at a very high level, with a large number clustered at 100% participation (the top horizontal line).
- ▶ The participation pattern is highly variable across all ages, especially between 40 and 80 years old.
- ▶ Middle-aged and older adults appear more likely to participate, while fewer younger members are present in the data, possibly because cancer diagnoses are less common at younger ages.
- ▶ Some members are recorded as age 0 or even -51, which may reflect data entry errors or missing age information.

Method: Linear Regression

We wanted to see whether age is related to participation, so we used a linear regression model. We made a scatterplot with age on the x-axis and participation on the y-axis, and added a line of best fit to see the overall trend.

R^2 : 0.003334697

- ▶ The R^2 value we got was 0.003, which means age only explains about 0.3% of the variation in participation. This is a really small number, so it suggests that age isn't a strong predictor.
- ▶ Although the relationship is weak, the plot shows that older members appear more engaged overall.
- ▶ This may be because cancer is more prevalent in older adults, who therefore make up a larger and more active portion of the community.

Research Question 3

Is the proportion of breast cancer diagnoses in male members equal to 0.2?

We test whether the proportion of male members diagnosed with breast cancer is 0.2% (i.e., 0.002). This allows us to evaluate whether the observed rate in Wellspring's data is consistent with a benchmark provided by the Canadian Cancer Society[2].

Motivation

Breast cancer is often associated with women, but men can also be diagnosed with it [2]. If male members in Wellspring's data are also affected, even at a low rate, it's important that support services are designed with them in mind. This question helps assess whether male breast cancer cases are present and in line with national expectations, so organizations like Wellspring can ensure their services are inclusive and evidence-based.

Data Cleaning

Table 4: Summary of Data Cleaning for Question 3

Reason for Removal	Number of Members Removed
Missing gender or breast info	1958
Non-Male members	2417

We filtered out 1,958 members who had missing gender or breast cancer information (Table 4).

Since this analysis focuses specifically on male breast cancer, we also removed all non-male records (2,417).

Therefore, we removed 4375 members in total.

Summary of Breast Cancer Incidence Among Males

We summarize the number and proportion of male members diagnosed with breast cancer (Table 5).

Table 5: Number and Proportion of Male Members Diagnosed with Breast Cancer

Breast Cancer Diagnosis	Number of Members	Proportion
FALSE	448	0.9867841
TRUE	6	0.0132159

- ▶ Among male members in the dataset, 6 out of 454 (or 1.3%) were diagnosed with breast cancer.
- ▶ This observed rate is much higher than the expected 0.2%, based on external data from the Canadian Cancer Society.
- ▶ In this dataset, approximately 1 in 76 male members were diagnosed with breast cancer.

Breast Cancer Diagnosis in Male Members

This bar chart provides a more direct visual comparison of breast cancer diagnosis rates between males and females (Figure 3).

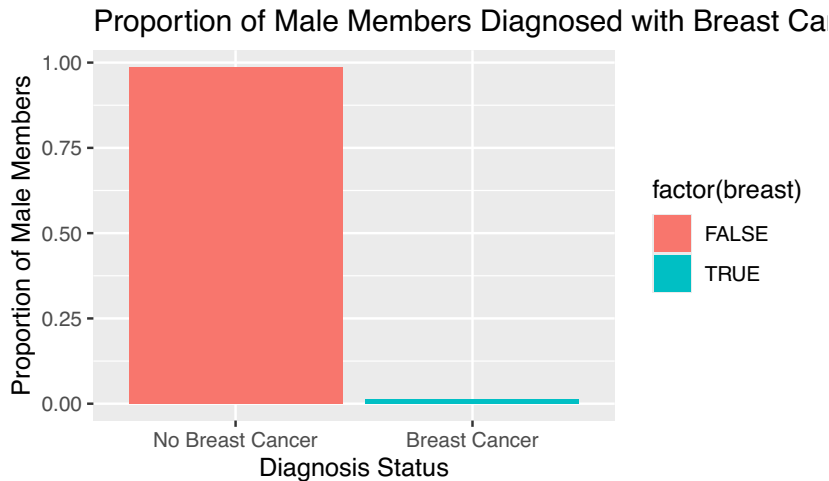


Figure 3: Proportion of Male Members Diagnosed with Breast Cancer

Method: Hypothesis test for one proportion

We tested:

- ▶ H_0 : The true proportion of male breast cancer diagnoses is 0.2%
- ▶ H_A : The true proportion is different from 0.2%
- 1. In each of 10,000 simulations, we randomly generated 454 male members under the assumption that only 0.2% would be diagnosed.
- 2. Then calculated how often the simulated proportions were as extreme or more extreme than the observed rate in our data (6 out of 454, or ~1.3%).

p-value: 1e-04

- ▶ The p-value was less than 0.001, meaning this observed result would be extremely unlikely if the true rate were 0.2%.
- ▶ The breast cancer rate among male Wellspring members (1.3%) is noticeably higher than the expected 0.2%. While this may reflect a small sample size or an extreme case, it raises important questions about whether current programs meet the needs of male patients. Wellspring may consider more targeted support for this group.

Limitations

1. March 2024 joiners were excluded because we didn't have exact registration dates, this reduces the sample size and may weaken our ability to detect the full impact of the new system.
2. Many records with missing values were removed, which could affect the results if the missing data followed a certain pattern.
3. Some age values (e.g., 0, -51) are unrealistic, which indicates possible data entry errors. Although we removed members with clearly invalid ages, some unrealistic values like 0 might still be in the data and could affect the accuracy of results about age and participation.
4. Some groups had small and uneven sample sizes, especially for male members in the breast cancer analysis. Since only 6 male members were diagnosed, the proportion is less stable and could either overstate or understate the true difference between genders.

Conclusion

- ▶ New members missed more events (31.4%) than old members (18.7%), suggesting that while the new sign-up system makes access easier, some new members may be less involved. Thus, Wellspring could consider extra support for new members, like reminder emails or onboarding.
- ▶ Older adults were more active overall, likely because they make up a larger portion of members. This suggests Wellspring could consider offering even more programs tailored to their needs.
- ▶ Among male members, 1.3% were diagnosed with breast cancer, which is much higher than the national average of 0.2%. While the number is small, it raises concerns about whether men with breast cancer are receiving enough attention and support in these programs. This raises questions about whether current services meet the needs of male patients.

Reference

- [1] Escalante, C. P. (2025). *Overview of psychosocial issues in the adult cancer survivor*. In: UpToDate, Wolters Kluwer.
<https://www-uptodate-com.myaccess.library.utoronto.ca/contents/cancer-related-fatigue-treatment>
- [2] Canadian Cancer Society. (November 2023). *Canadian Cancer Statistics: A 2023 special report on cancer prevalence*. ISSN 0835-2976.
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